Gender Classification Using PCA, LDA, and k-NN

1. Task Description

This project implements a biometric system for gender classification using facial image data. The goal is to distinguish between male and female individuals by analyzing their facial features. The system employs classical pattern recognition techniques such as Principal Component Analysis (PCA) and Linear Discriminant Analysis (LDA) for feature extraction and dimensionality reduction. These are followed by classification using a custom implementation of the k-Nearest Neighbors (k-NN) algorithm. The purpose is to evaluate the effectiveness of these techniques in a controlled biometric environment while ensuring the system remains interpretable and computationally efficient.

2. Methodology

2.1 Preprocessing

The input dataset consists of facial images which are first converted to grayscale and uniformly resized to 128×128 pixels to maintain consistency across all samples. To enhance the quality of local image features, the Contrast-Limited Adaptive Histogram Equalization (CLAHE) technique is applied, which improves contrast and helps in better capturing subtle facial characteristics. The processed images are then flattened into one-dimensional column vectors, creating a matrix suitable for further feature extraction processes.An 80:20 train-test split is applied to prepare the data for feature extraction and classification.

2.2 Feature Extraction

For dimensionality reduction, the first step is the application of Principal Component Analysis (PCA). PCA identifies the directions of maximum variance in the dataset and projects the original data onto a lower-dimensional subspace. In the PCA-only classification pipeline, the number of principal components is empirically chosen based on experimental tuning for each dataset size. For example, with 100 images, 100 components are used; with 200 images, 75 components; and with 1000 images, 80 components are selected.

In the PCA → LDA pipeline, PCA first reduces the feature space to a lower dimension before applying LDA. Specifically, 30 PCA components are used for 1000 images, 15 for 200 images, and 30 for 100 images. These values are chosen to maintain a balance between retaining important variance and avoiding overfitting in high-dimensional spaces, especially given the varying size of the datasets.

2.3 Classification

Following feature extraction, classification is performed using a kFollowing feature extraction, classification is performed using a k-Nearest Neighbors (k-NN) algorithm. Rather than using a fixed value, the parameter k can be tuned based on the size and complexity of the dataset to optimize performance. For smaller datasets, such as the 100 and 200-image experiments, k=3

was used, while in the 1000-image experiment, k=5 was empirically found to give strong results. Unlike traditional Euclidean distance, cosine similarity is employed to compute distances in the high-dimensional feature space. Cosine similarity is particularly effective here, as it emphasizes angular differences between vectors, which tends to yield better performance with normalized image data. Two classification pipelines are tested: PCA + k-NN, and PCA followed by LDA + k-NN.

3. Results and Evaluation

3.1 Classification Accuracy Across Dataset Sizes

The classification accuracy of both methods was tested across datasets of varying sizes: 100, 200, and 1000 images. The performance results are summarized in Table 1.

Method	**100 Images** (50 female, 50 male)	**200 Images** (100 female, 100 male)	**1000 Images** (500 female, 500 male)
PCA + k-NN	95.0%	82.5%	81.5%
PCA $\rightarrow$ LDA + k-NN	90.0%	92.5%	81.5%

Table 1: Classification accuracy for different dataset sizes.

Accuracy is significantly affected by the values of K_pca_main, k_pca_lda, and k. These parameters define the number of principal components retained, the dimensionality fed into LDA, and the number of neighbors considered during classification. Optimal tuning of these values is essential for maximizing performance, especially as the size and variance of the dataset increase.

3.2 Confusion Matrices

```
Confusion Matrix (PCA):
    12     1
     0     7

Confusion Matrix (LDA):
    13     0
     2     5
```

Figure 1: Confusion Matrix for 100 images

```
Confusion Matrix (PCA):
    15     2
     5    18

Confusion Matrix (LDA):
    16     1
     2    21
```

Figure 2: Confusion Matrix for 200 images

```
Confusion Matrix (PCA):
    83    23
    14    80

Confusion Matrix (LDA):
    87    19
    18    76
```

Figure 3: Confusion Matrix for 1000 images

3.3 ROC Curve

The ROC curves below compare the performance of the PCA + k-NN and PCA → LDA + k-NN classifiers. The LDA-based model generally achieved higher AUC values, indicating stronger discriminative capability.

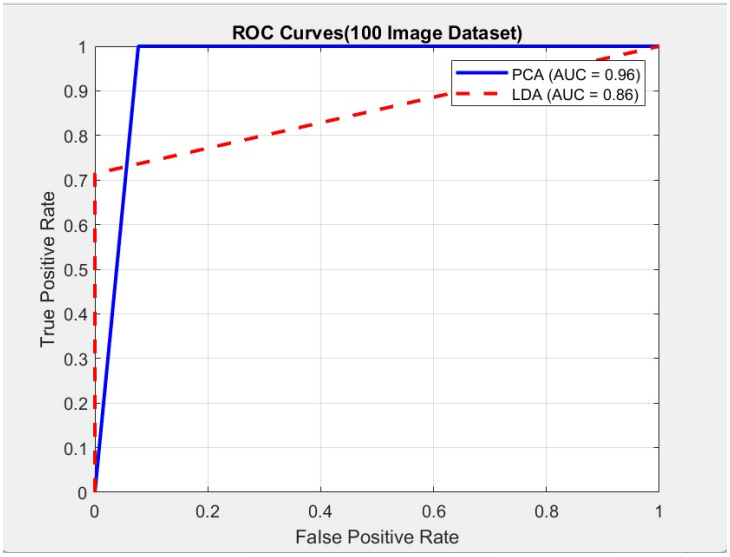


Figure 4: ROC Curve with 100 Images

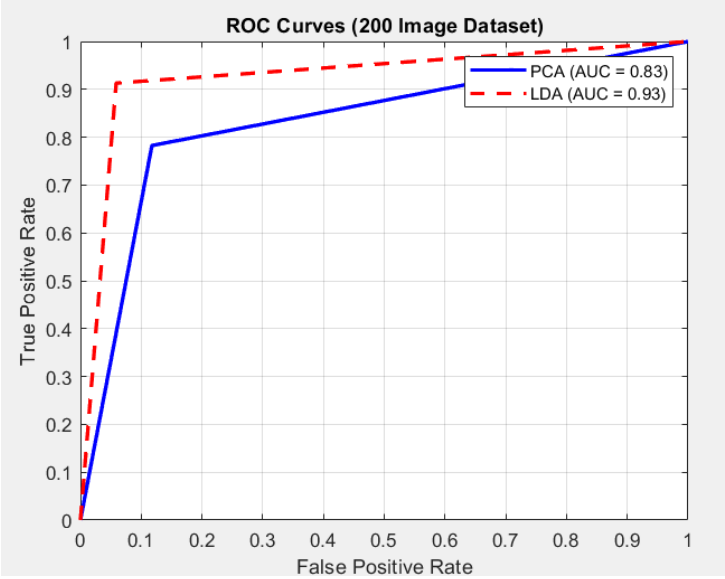


Figure 5: ROC Curve with 200 Images

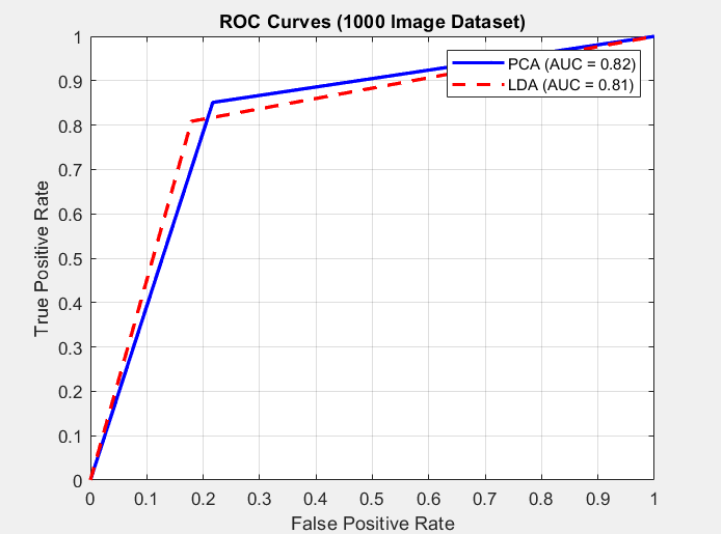


Figure 6: ROC Curve with 1000 Images

4. Observations

The PCA-based classification pipeline successfully captures variability in facial data, offering moderate performance. However, its performance is improved when followed by LDA, which enhances class separability by focusing on inter-class differences. The use of cosine similarity further improves results in high-dimensional feature spaces compared to Euclidean distance. Importantly, the confusion matrices reveal balanced performance across gender classes, indicating the absence of strong classification bias. These observations suggest that the combined PCA–LDA pipeline is a reliable approach for lightweight, interpretable biometric systems.

5. Conclusion

Integrating PCA and LDA for dimensionality reduction with a cosine-based k-Nearest Neighbors classifier offers an effective solution for gender classification from facial images. The approach achieves consistent and balanced accuracy across varying dataset sizes while maintaining computational efficiency. Its simplicity and interpretability make it a practical choice for applications where deep learning methods are constrained by resource or data limitations. These results emphasize the enduring strength of classical machine learning techniques in real-world biometric systems.